User Manual – Project Nilbye Prototype

A collaboration between Welthungerhilfe and Werk:Raum

Nilbye.

werk:raum.thws

Startup-Lab im Institut für Digital Engineering

Technische Hochschule Würzburg-Schweinfurt

Manual Content

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API	Application Programming Interface
DC	Direct Current
DHCP	Dynamic Host Configuration Protocol
GUI	Graphical User Interface
GPIO	General Purpose Input/Output
HDMI	High-Definition Multimedia Interface
HTTP	Hypertext Transfer Protocol
IP	Internet Protocol
LTE	Long Term Evolution
MPPT	Maximum Power Point Tracking
MQTT	Message Queuing Telemetry Transport
PTZ	Pan-Tilt-Zoom
RTSP	Real-Time Streaming Protocol
SSH	Secure Shell
USB	Universal Serial Bus
VPN	Virtual Private Network
YOLO	You Only Look Once

Introduction

The Project Nilbye first-stage prototype is an experimental device developed by the Werk:Raum team in collaboration with Welthungerhilfe. The prototype uses open-source software and a DeepStream pipeline to deploy both trained and pre-trained YOLO models via RTSP with a PTZ camera, all on the versatile, compact NVIDIA Jetson Orin Nano. This pipeline enables real-time object detection, all controlled smoothly via a customizable API that supports a graphical user interface (GUI).

As for the hardware, many electronic components and devices were procured to set up the required solar energy network for field testing and deployment. In addition, electronics were connected to provide proper input for the detected images, in the form of ultrasound, which were activated by signals from the Jetson and communicated via the API using MQTT. The aforementioned energy network is meant to power the devices and components placed on the main case and the PTZ camera.

The complete system essentially has two arrays for input and output. The first is for the pipeline: the camera receives images as input, and the output is an ultrasonic frequency. The GPIO Header from the Jetson bridges these two points; see PHASE I. The second one is solar energy as input; see PHASE III, which is converted to DC for two outputs.

WARNING and **NOTE**

Warnings and Notes are critical signs placed in key locations throughout the manual to draw the user´s attention to both safety concerns and essential steps for the proper functioning of the prototype. Be sure to comply with all highlighted warnings and notes to ensure a successful setup and general safety.

NOTE: Due to limitations at the Werk:Raum facilities, the prototype could not be tested under all possible operating conditions. Therefore, this instruction set comprises three distinct phases: developer mode, headless mode, and finally integration with the energy network. This procedure is laid out to test the prototype in phases due to energy-consumption constraints. The aim for the future development of this Prototype is to streamline this entire process, so fewer instructions are needed.

WARNING: When connected, all devices and components should be constantly monitored. The system should never be left to run unsupervised. Failure to comply with these requirements could lead to severe damage to the components. The main case shown in the prototype description is 3D-printed. While not fragile, it can't withstand severe stress and is not completely waterproof. It is recommended to handle all components with extreme care and, if possible, avoid exposure to extreme conditions without proper supervision. If necessary, the lid can be lifted off to prevent overheating. Not all necessary loads on the energy network were fully tested, so it is advisable to take extra precautions in the final steps, especially in PHASE III.

The following items are all the devices used in the system. Figures 1 to 10 show their main specifications

Prototype Description

NVIDIA Jetson Orin Nano (Includes power cable).

1. TP-Link 4G LTE Mobile Wi-Fi (Includes Micro USB cable).
2. Litionite Venom Powerbank.
3. Kemo Ultrasonic High-Power Generator (Includes Jack cable).
4. Songle SRD-05VDC-SL-C Relay with respective connections.
5. Dahua PTZ Camera (Includes power cable).
6. Enjoy Solar Panel.
7. Victron SmartSolar MPPT 75|15.
8. Greencell 12 V Battery.
9. Voltcraft SW-150 Inverter.

Additional components, as shown in Figure 58:

1. Battery connectors.
2. DIN rail.
3. 0.75 mm² and 4 mm² cables.
4. 15 Amp fuse with a 4 mm^2 respective insert cable.
5. Wago 221 connectors.

System Specifications

Architecture	Embedded AI system with external PTZ camera and ultrasonic deterrence
Processing unit	NVIDIA Jetson Orin Nano
Detection method	Vision-based object detection (YOLO via DeepStream)
Camera input	RTSP stream from PTZ camera
Deterrence output	Ultrasonic signal via relay-controlled generator
Control interface	REST API with web-based GUI
Primary system voltage	12 V DC
Primary power source	Solar panel with battery storage
Secondary power source	Litionite Venom Powerbank
Power conversion	DC–DC and DC–AC via inverter (Phase III)
Typical power consumption	Approx. 20–30 W
Maximum power consumption	Approx. 35–50 W
Operating modes	Developer mode / Headless mode / Energy network mode
Network connectivity	Ethernet, Wi-Fi, 4G LTE
Communication protocols	RTSP, HTTP, MQTT, SSH
Cooling concept	Passive cooling
Deployment type	Stationary prototype, supervised operation
Environmental operating temperature	−10 °C to +50 °C
Intended use	Experimental prototype for real-time detection and deterrence

NVIDIA Jetson Orin Nano Developer Kit (8 GB)

Figure 1. NVIDIA Jetson Orin Nano Developer Kit

Supply voltage	5–20 V DC (via DC jack or carrier board input)
Typical operating voltage	12 V DC
Power consumption	7 W (low power mode) – 15 W (max)
CPU	6-core Arm Cortex-A78AE
GPU	NVIDIA Ampere, 1024 CUDA cores, 32 Tensor cores
AI performance	Up to 67 TOPS
Memory	8 GB LPDDR5
Storage	microSD, NVMe (M.2 Key-M)
I/O interfaces	USB 3.2, HDMI, Ethernet, 40-pin GPIO
Networking	Gigabit Ethernet
Operating temperature	0 °C to +60 °C
Cooling	Passive heatsink (active optional)
Operating system	Ubuntu Linux (JetPack 6.x)

TP-Link 4G LTE Mobile Wi-Fi Router (M7/M7200 class)

Figure 2. TP-Link 4G LTE Mobile Wi-Fi Router

Power input	5 V DC (Micro-USB)
Current consumption	up to 1 A
Battery Internal Li-ion	approx. 2000 mAh
Charging source	USB (power bank or adapter)
Network standard	4G LTE
Wi-Fi standard	IEEE 802.11 b/g/n
Operating temperature	0 °C to +40 °C
Typical power consumption	2~3–5 W

Litionite Venom Power Bank (DC version)

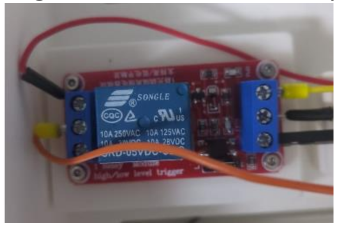
Figure 3. Litionite Venom Power Bank

Battery chemistry	Lithium-ion
Nominal battery voltage	11.1–12 V
Capacity	90–150 Wh (model dependent)
DC outputs	12 V / 16 V / 20 V / 24 V
Max output current (12 V)	up to 10 A
USB outputs	5 V DC
Input (charging)	DC IN
Protection	Over-current, over-voltage, short-circuit
Operating temperature	−10 °C to +45 °C

Kemo Ultrasonic High-Power Generator

Figure 4. Kemo Ultrasonic High-Power Generator

Supply voltage	12–15 V DC
Current consumption	500–800 mA
Power consumption	6–10 W
Output frequency	20–40 kHz
Frequency adjustment	Potentiometer
Output signal	Ultrasonic
Connection	3.5 mm jack
Operating temperature	−10 °C to +40 °C
Supply voltage	12–15 V DC
Current consumption	500–800 mA

Songle SRD-05VDC-SL-C Relay  Figure 5. Songle SRD-05VDC-SL-C Relay	Coil voltage	5 V DC
	Coil current	Approx. 70 mA
	Contact type	SPDT
	Max switching voltage	250 VAC / 30 V DC
	Max switching current	10 A
	Control method	GPIO-controlled
	Isolation	Electromechanical
	Operating temperature	−40 °C to +85 °C
	Coil voltage	5 V DC
	Coil current	Approx. 70 mA
Dahua PTZ Camera  Figure 6. Dahua PTZ Camera	Supply voltage	12 V DC or PoE
	Power consumption (idle)	Approx. 8 W
	Power consumption (active)	Up to 18 W
	Resolution	1920 × 1080
	Video compression	H.264 / H.265
	Streaming protocol	RTSP
	Network interface	Ethernet
	PTZ control	ONVIF / HTTP
	Operating temperature	−30 °C to +60 °C
	Supply voltage	12 V DC or PoE

Enjoy Solar Panel

Figure 7. Enjoy Solar Panel

Nominal system voltage	12 V
Open-circuit voltage (Voc)	Approx. 20–22 V
Maximum power voltage (Vmp)	Approx. 18 V
Rated power	50–100 W (model dependent)
Output type	DC
Operating temperature	−40 °C to +85 °C
Nominal system voltage	12 V
Open-circuit voltage (Voc)	Approx. 20–22 V
Maximum power voltage (Vmp)	Approx. 18 V
Rated power	50–100 W (model dependent)

Victron SmartSolar MPPT 75|15

Figure 8. Victron SmartSolar MPPT 75|15

Maximum PV input voltage	75 V
Maximum charge current	15 A
Battery voltage	12 V / 24 V (auto-select)
Maximum PV power (12 V system)	220 W
Conversion efficiency	>98 %
Communication	Bluetooth (VictronConnect)
Operating temperature	−30 °C to +60 °C
Maximum PV input voltage	75 V
Maximum charge current	15 A
Battery voltage	12 V / 24 V (auto-select)

Greencell 12 V Battery

Figure 9. Greencell 12 V Battery

Nominal voltage	12 V
Battery chemistry	AGM / Li-ion (model dependent)
Capacity	20–50 Ah
Energy content	240–600 Wh
Maximum discharge current	20–40 A
Operating temperature	−20 °C to +50 °C

Voltcraft SW-150 Inverter

Figure 10. Voltcraft SW-150 Inverter

Input voltage	12 V DC
Output voltage	230 V AC
Rated power	150 W
Peak power	Approx. 300 W
Output waveform	Modified sine wave
Efficiency	Approx. 85 %

The 3D- Printed case itself has four levels. Figure 11 shows the assembled prototype main case containing all listed components. The prototype enclosure contains the following electronic components and peripherals. The internal structure of the case is illustrated in Figures 11 to 15. The hexagonal screws for bolting the levels together are provided:

Figure 11. Prototype Main Case

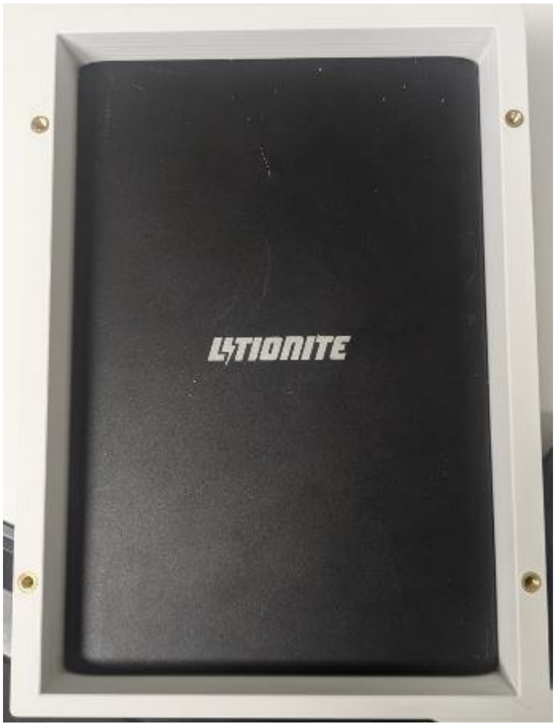


Figure 12. Level 0 with the powerbank inserted

Figure 13. Level 1 with both mounted devices

Figure 14. Relay with circuit

Figure 15. Lid that encloses the main case

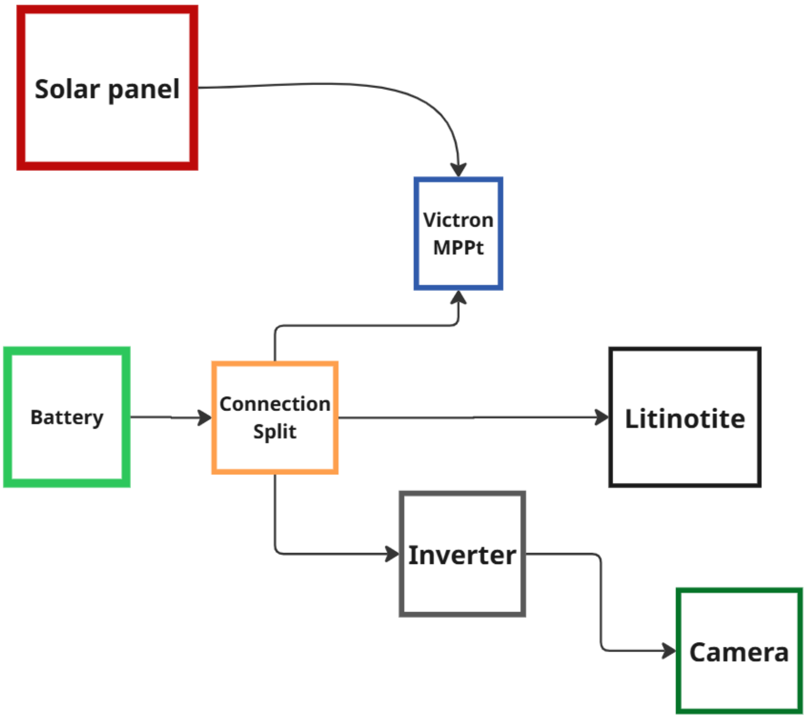


Figure 16. Complete energy network

PHASE I: Developer Mode

In this stage, it is required to configure network connectivity so that the Prototype´s API executes the DeepStream pipeline, enabling real-time object detection via RTSP as the system´s input while simultaneously triggering signals to activate the ultrasound, which serves as the system's output.

NOTE: At this point, the system should be powered by a fixed power source and not to be connected directly to the Litionite Venom Power bank or the energy network. This measure is taken to ensure more efficient energy consumption while the user configures the necessary networks and adjusts the required documents for the pipeline to function.

WARNING: Do not skip these steps; connect this device to the power bank and the energy network only when indicated.

1. Setup the TP-Link 4G LTE Mobile Wi-Fi
    - 1.1. Unbox the device
    - 1.2. Remove the lid
    - 1.3. Remove the battery
    - 1.4. Place the SD-Card in the black sleeve
    - 1.5. Insert the sleeve with the card into the slot
    - 1.6. On the device is the network name and password. Write this for later.
    - 1.7. Place the battery and the lid back
    - 1.8. Turn on the device; its initial configuration state is shown in Figure 12
    - 1.9. On a PC or laptop running Windows connected to the network TP-Link_B8E6
    - 1.10. Insert the password 52846515 to connect to the network
    - 1.11. Launch a browser and visit http://tplinkmifi.net or http://192.168.0.1
    - 1.12. Create a new username or password and then log in. The login window and dashboard are shown in Figures 17 and 18.

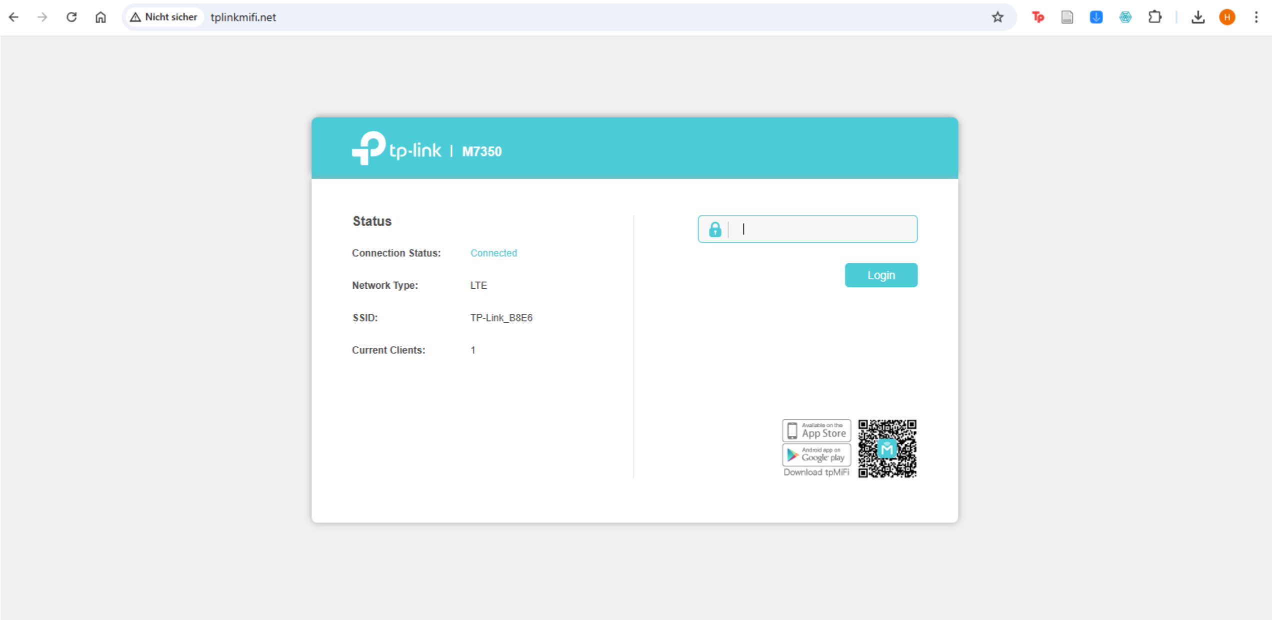


Figure 17. Tp-link log-in window

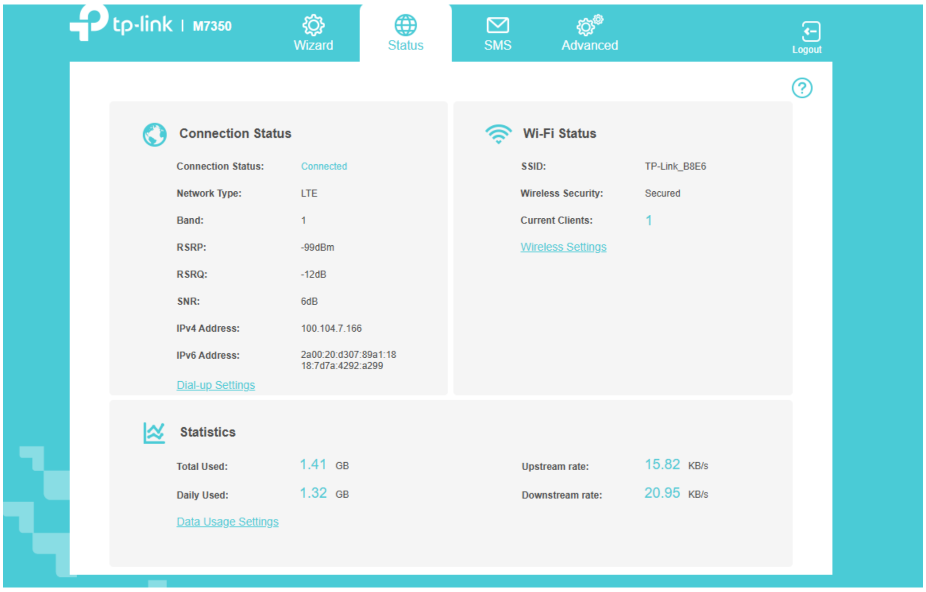


Figure 18. Tp link Dashboard

2. Setup the NVIDIA Jetson Orin Nano

2.1. Unbox the device, including the power cable. Grab the device and bolt it with holder 1 on Level 1 . Figure 19 shows the mounted Jetson Orin Nano inside the case.

Figure 19. Mounted Jetson Close-up

2.2. The following components are needed for the next step. The available connection interfaces are shown in Figure 20:

- HDMI Cable
- Mouse with USB connection
- Ethernet Cable
- Keyboard with USB connection

2.3. Connect them in their respective slots on the Jetson as shown in the image below.

Figure 20. Jetson connection slots for peripherals

2.4. Bolt the TP-Link 4G LTE Mobile Wi-Fi with holder 2.

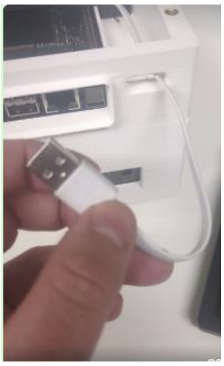


Figure 21. Tp link mirco usb cable

Figure 22. Mounted tp-link

2.5. Connect the cable to the loading port as seen in the image and get it through the slot on Level 1. Leave it hanging for now.

3. Connect the Songle SRD-05VDC-SL-C Relay to NVIDIA Jetson Orin Nano.
   3.1. Unbolt the lid from Level 2 to separate both. Gently connect female IDC ribbon cable connector to the 40 GPIO Header. The cable should be connected to PIN 6 (GND) and the yellow to PIN 7. Figure 23 illustrates the GPIO connections used for the relay control.

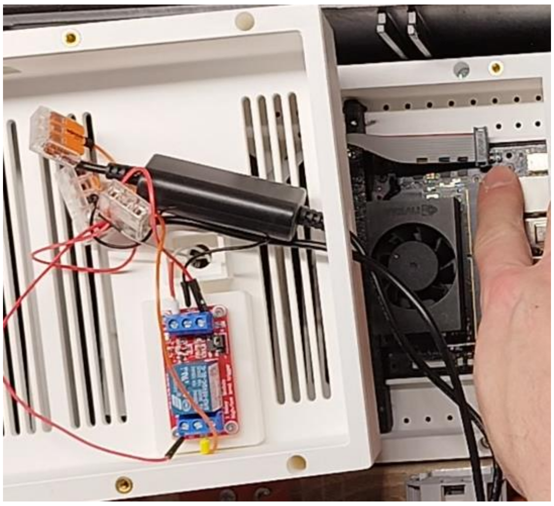


Figure 23. GPIO connections with IDC cable

3.2. Bolt Level 2 to Level 1 with the square slot aligned with the NVIDIA. The image below shows an example of the 3 main levels mounted together. For the final deployment this array won´t be necessary but it can be useful for isolating testing with fix power.

Figure 24. Example of Level 0, Level 1, and Level 2 are mounted together

3.3. Get the USB cable and connect it to the respective slot on the Jetson.

For now, the ultrasound won´t be tested but the response to the detections can be identified on the relay itself when a red-light displays as seen on the example below.

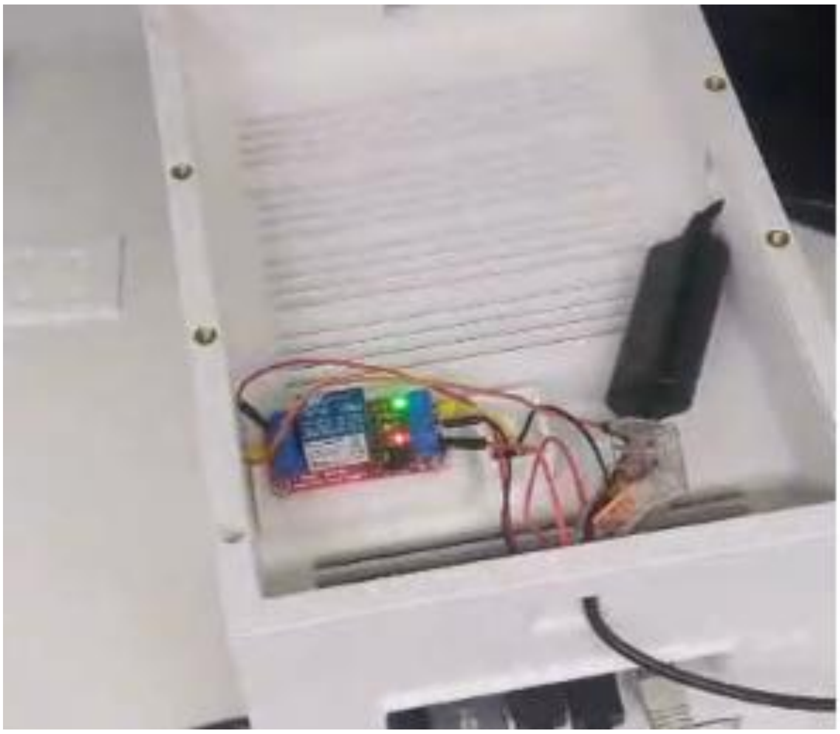


Figure 25. Relay respond with detection

4. Power the PTZ Camera
    4.1. Unbox the camera
    4.2. Bolt the holder piece to the main body
    4.3. Bolt the holder to a surface as shown in the following examples. Suggested exterior and interior mounting positions are shown in Figures 26 and 27.

Figure 26. Propose use for the exterior at around 2 to 3 meters from the ground

Figure 27. Propose use for the interior at sitting level

    4.4. Grab the charger from the PTZ Camera. Connect the negative cable (black) with a Wago connector to the negative cable (black) coming from the camera. Do the same with the red cables. The Ethernet cable connected to the Jetson should be plugged into the camera's Ethernet slot. Figures 28 to 30 show the PTZ camera power and wiring connections.

4.5. Connect the charger to a fixed power source. **NOTE:** At this point, the camera is connected to a fixed power source, not directly to the energy network. This measure is taken to ensure a way to allow for early demonstrations and avoid problems. **WARNING:** Do not skip this step; connect this device to the energy network only when indicated.

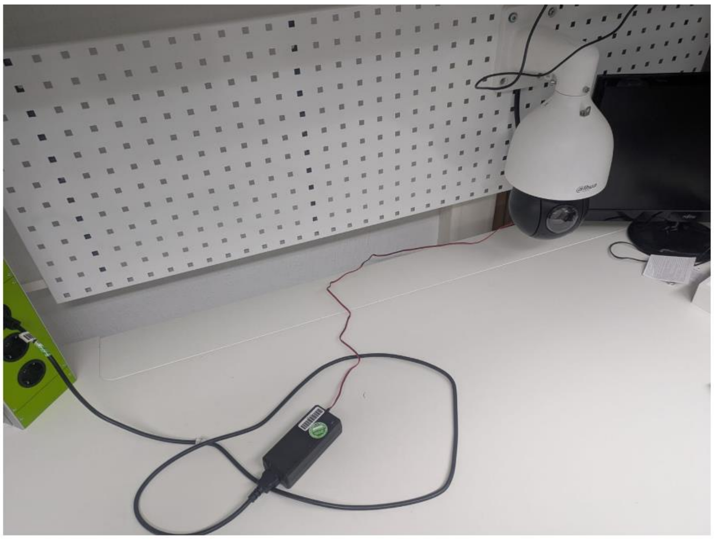


Figure 28. Plug-in PTZ camera

Figure 29. Camera charger

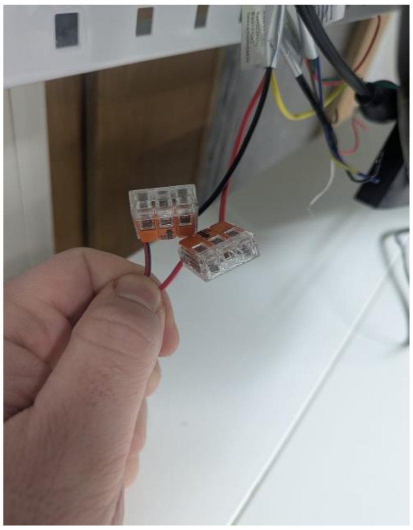


Figure 30. Positive and negative wires

4.6. Wait around 5 min for the camera to boot

5. Power up the Jetson and start the API

5.1. Connect the charger to a fixed power source

5.2. Connect the jack to the NVIDIA. It should power up. The Jetson power connection sequence is shown in Figures 31 to 33.

Figure 31. Connecting the Jetson to a fixed power

Figure 32. Jetson charge cable

Figure 33. Close up to connection

5.3. The login should be automatic. It would take around 3 min. Figures 34 to 36 show the Jetson startup and browser authentication.

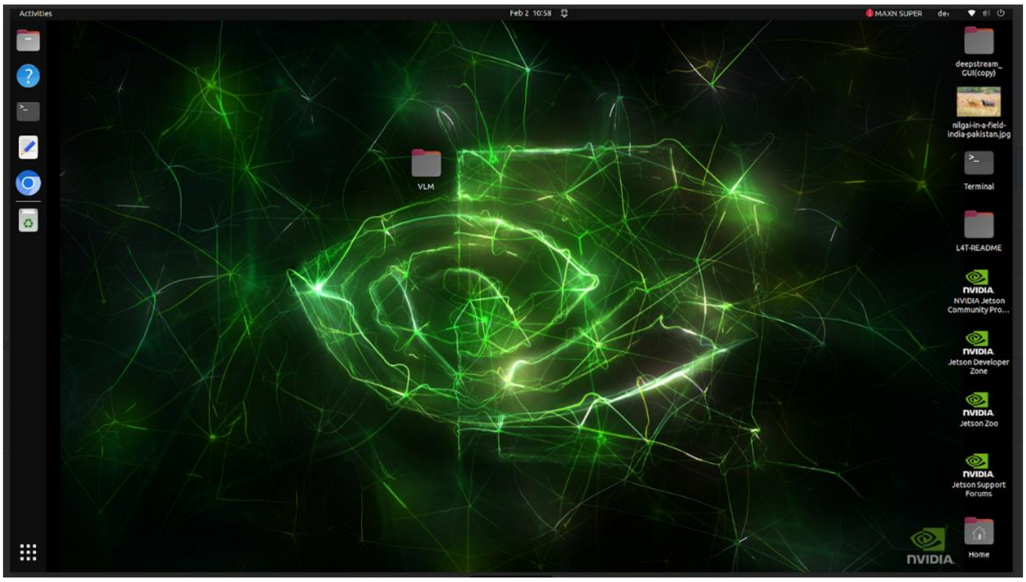


Figure 34. Jetson main screen

5.4. Once logged in, the upper right side, connect to the TP-Link Wi-Fi network with the same password as in step 1.10.

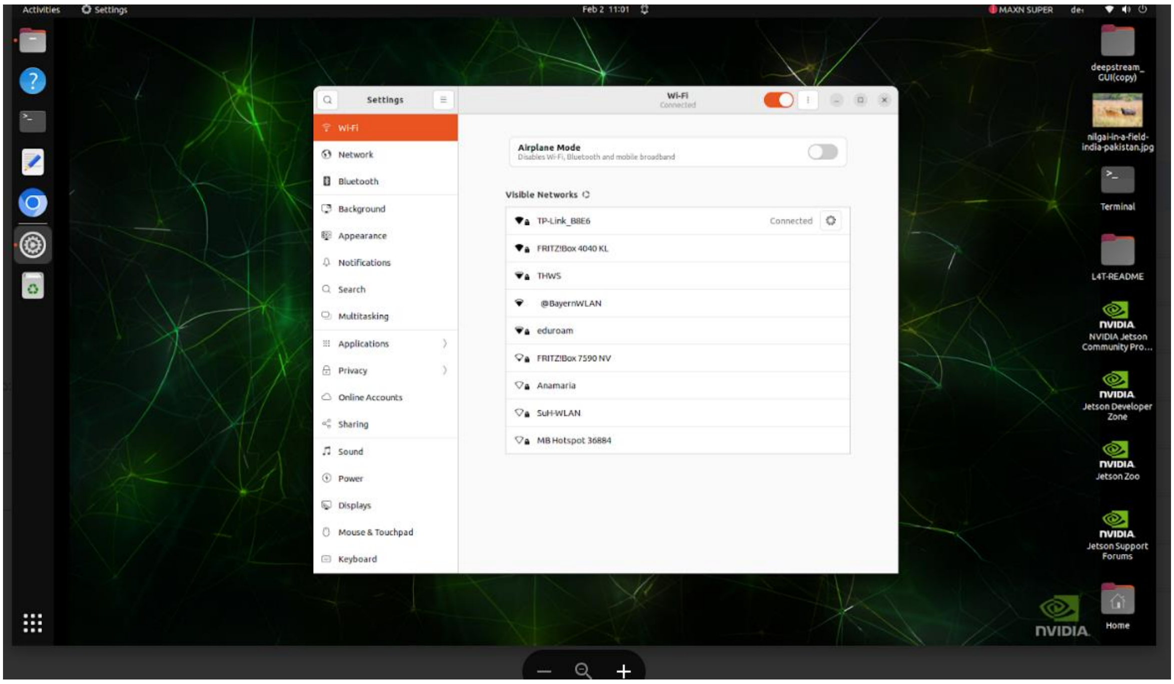


Figure 35. Jetson Wi-Fi connection

5.5. Click on the Chromium browser on the left of the screen and insert the same login password dead_wild. The browser should stay opened.

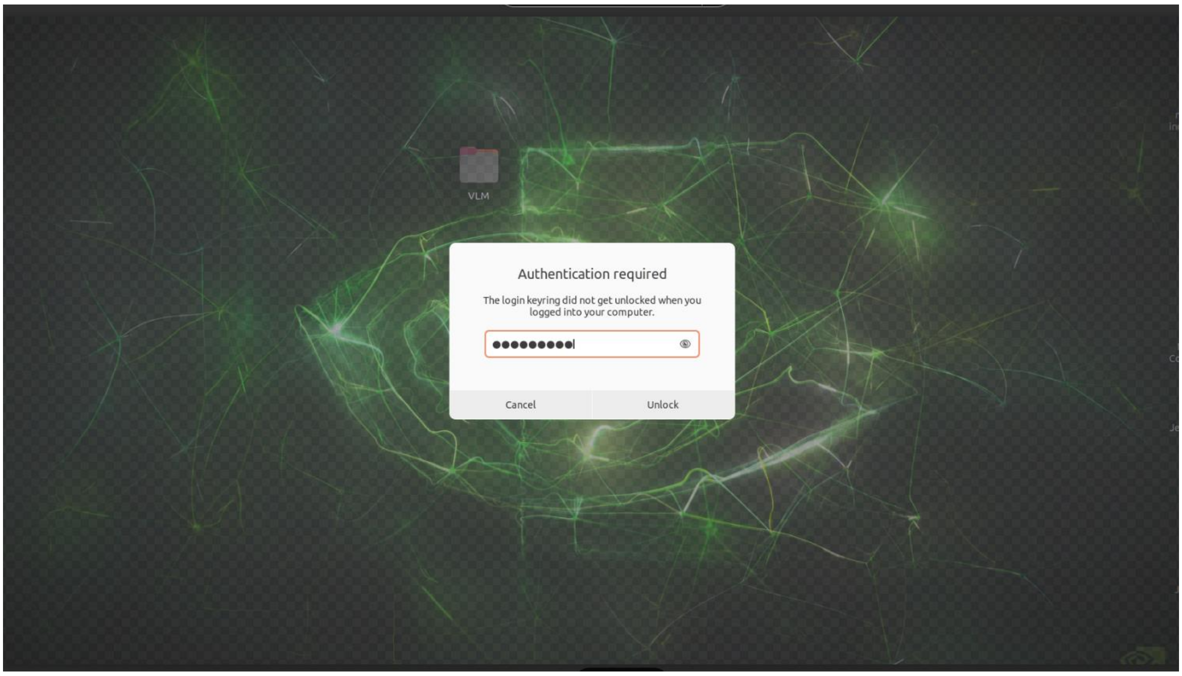


Figure 36. Browser authentication

5.6. In the Home folder, click on DeepStream-Yolo-master. In this folder, click on a file called Deepstream_example_usb.txt. The home directory is displayed in Figure 37.

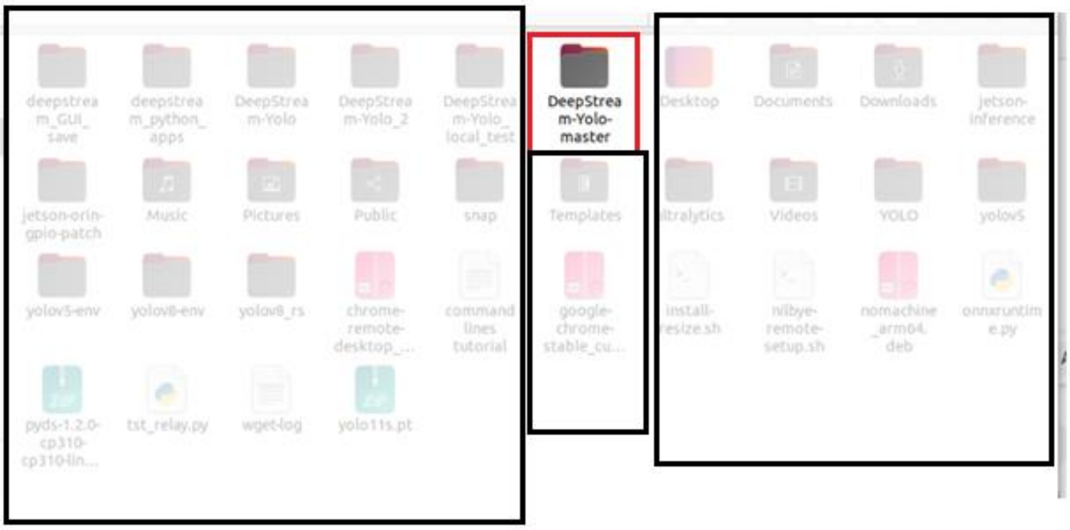


Figure 37. Home directory

So far, the user should be able to access and control the Dahua PTZ camera and execute the pipeline using the API as outlined in the next steps. This is because the camera is pre-configured to use automatic IP assignment (DHCP). If this configuration fails due to technical difficulties, please use this fallback method. If the user successfully executes this configuration, continue to step 5.7.

Fallback Method:

1. Remove the Ethernet cable from the Jetson
2. Connect the cable to a router with a LAN port
3. On a pc or laptop with Windows, download the ConfigTool from Dahua´s official site https://dahuawiki.com/ConfigTool.
4. Install the ConfigTool. Make sure the pc or laptop is connected to the same network as the router.
5. Run the application as administrator.
6. On the dashboard, as seen in this example, press search settings, and in the password field, write krishna_01. Leave the username as admin. Then, click on refresh, and the camera should appear as initialized. It should be similar to Figure 38.

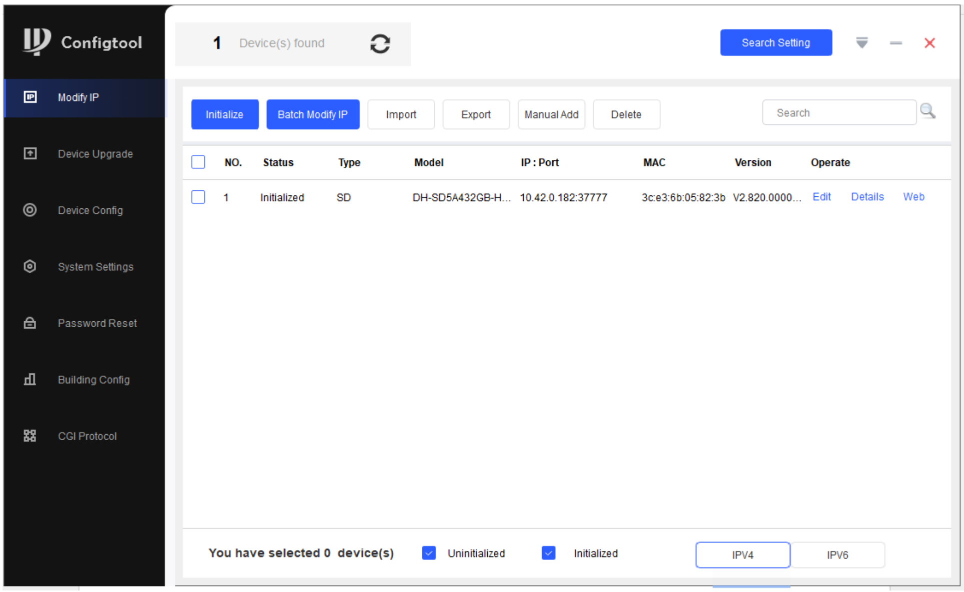


Figure 38. ConfigTool device list

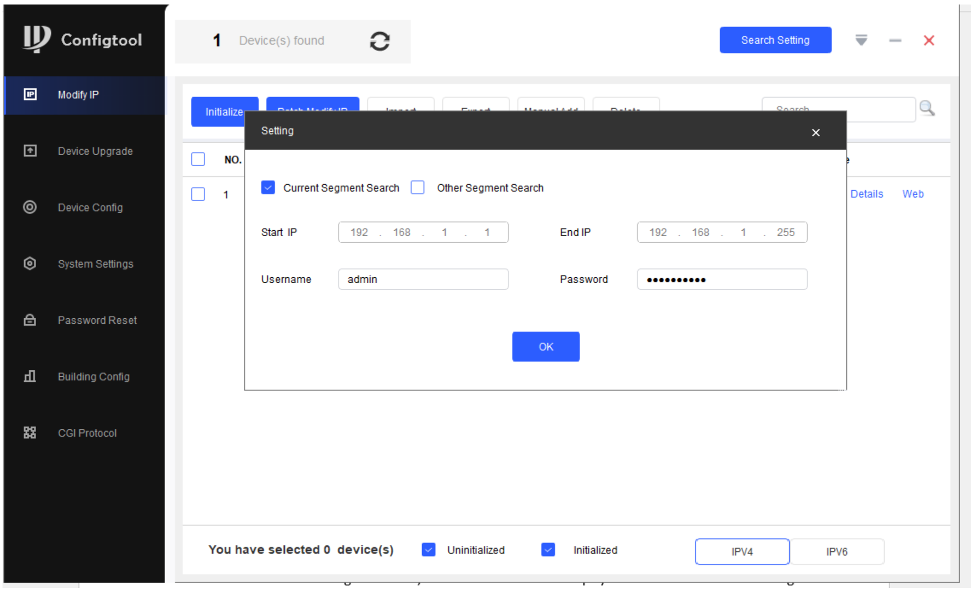


Figure 39. Batch Modify IP window

7. Select the camera on the square and press on Batch Modify IP as seen in Figure 39.

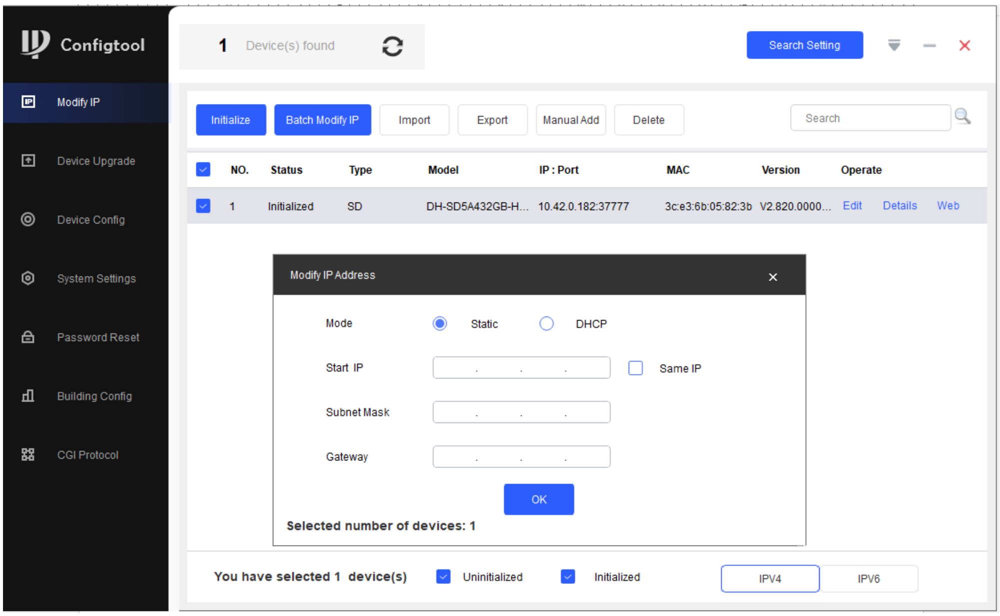


Figure 40. Static vs DHCP

8. Press DHCP to automatically use the network. Click on the circle shown on Figure 40.
9. Click OK.
10. Wait between 10 and 20 seconds
11. Click Search again. The camera should be listed
12. If this previous step wasn´t successful, manual IP configuration is possible. On step 7, instead of DHCP, press Static.
13. Change the fields with the following information from a working configuration. After filling these fields, repeat steps 9,10 and 11.

Camera IP: 10.42.0.182
Subnet Mask: 255.255.255.0
Gateway: 10.42.0.1

14. Return to step 5.6 and follow the rest of the instructions from there.

5.7. On DeepStream-Yolo-Master folder Click on the deepstream_GUI folder as shown in Figure 41.

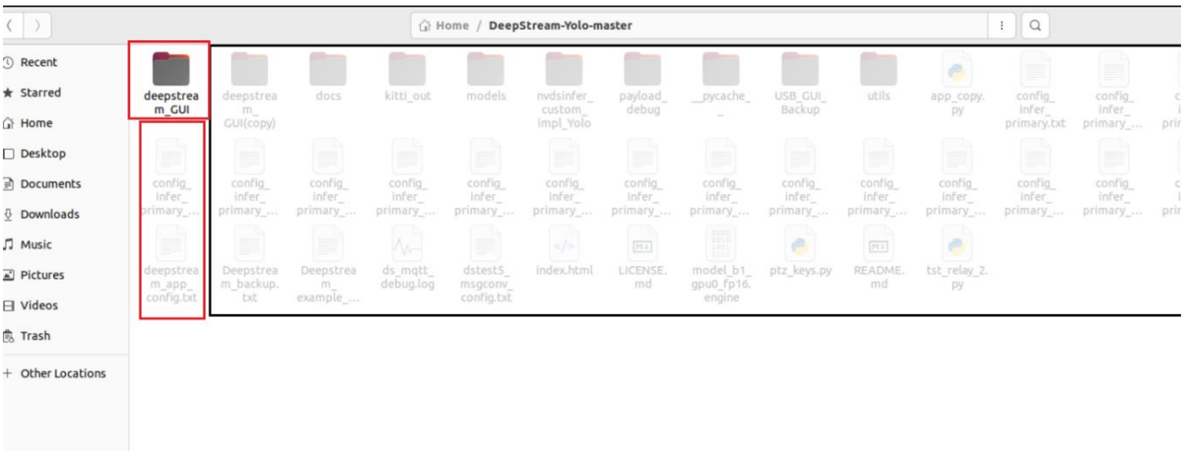


Figure 41. DeepStream folder

5.8. While in the deepstream_GUI folder, click on the three points on the upper right as shown in Figure 42.

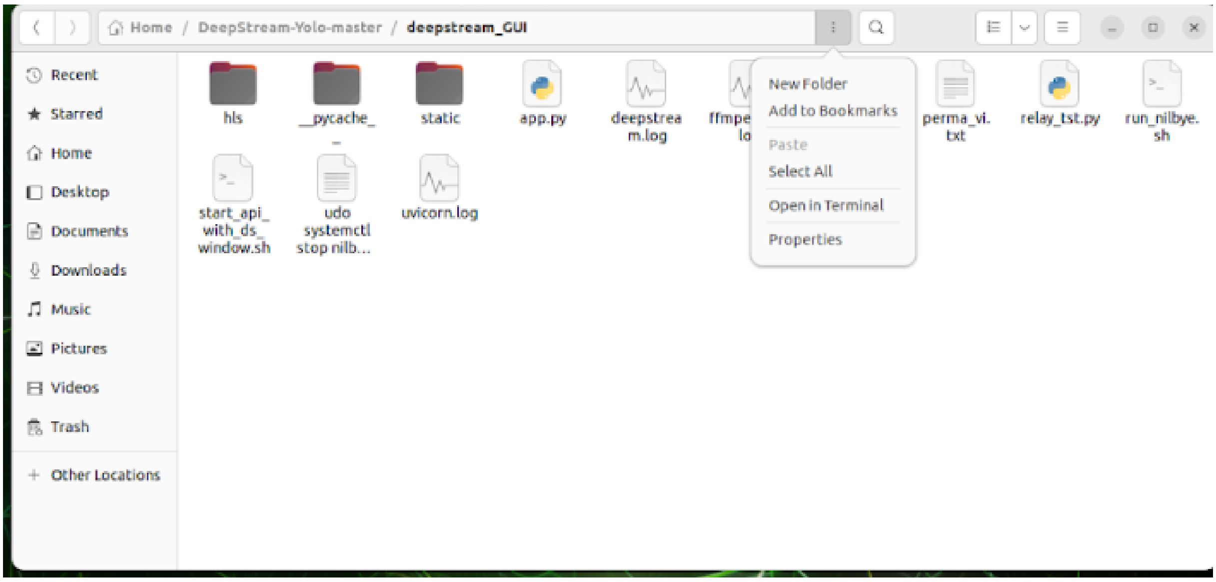


Figure 42. GUI folder

5.9. Open a new terminal from this folder and insert the following command: uvicorn app: app --host 0.0.0.0 --port 8000. Similar messages should be seen as in Figure 43.

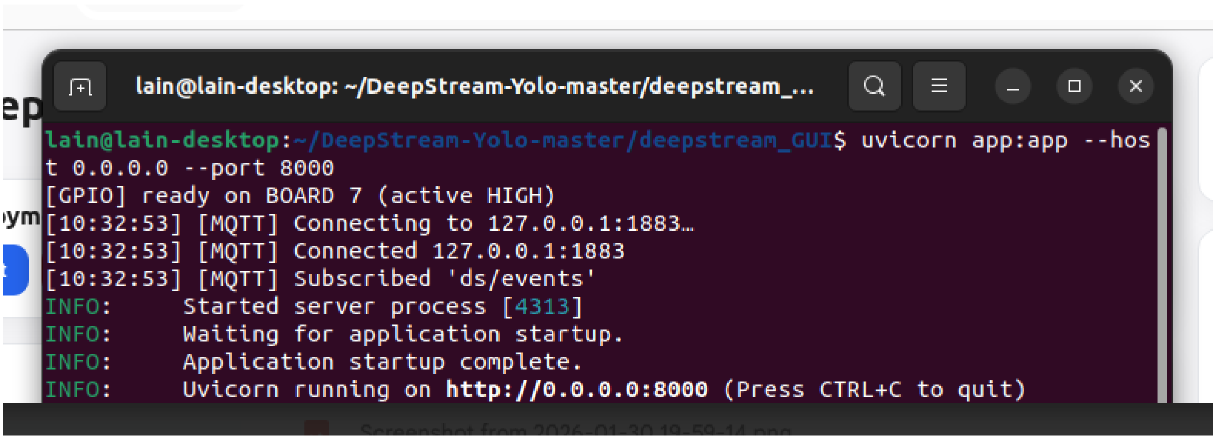


Figure 43. Uvicorn with terminal

5.10. Enter the URL shown in the terminal, or right-click http://0.0.0.0:8000 and open the link.

5.11. The GUI will appear on the screen as seen in Figure 44. Press Start. A window should pop up on the screen. The model uses around 80 classes, including some animals. This can be used to detect many objects, including people and office items such as laptops and keyboards.

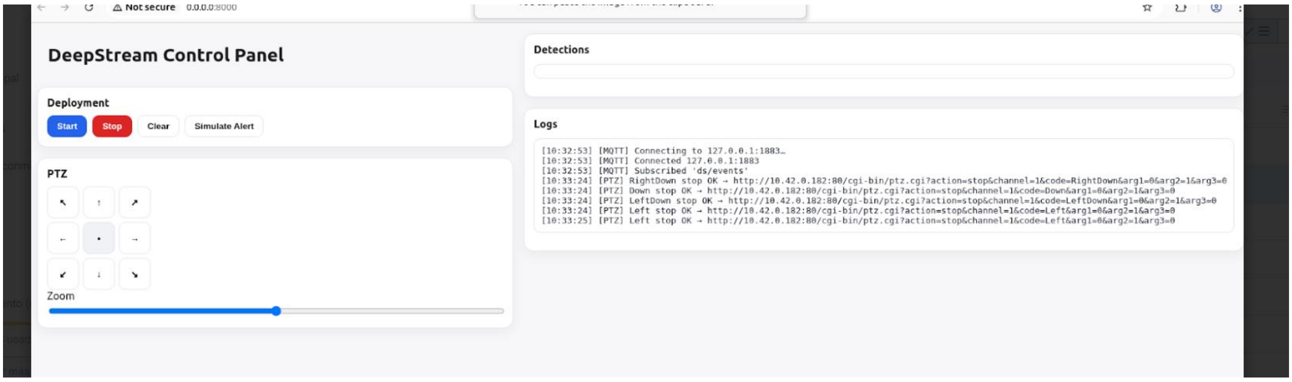


Figure 44. GUI control panel

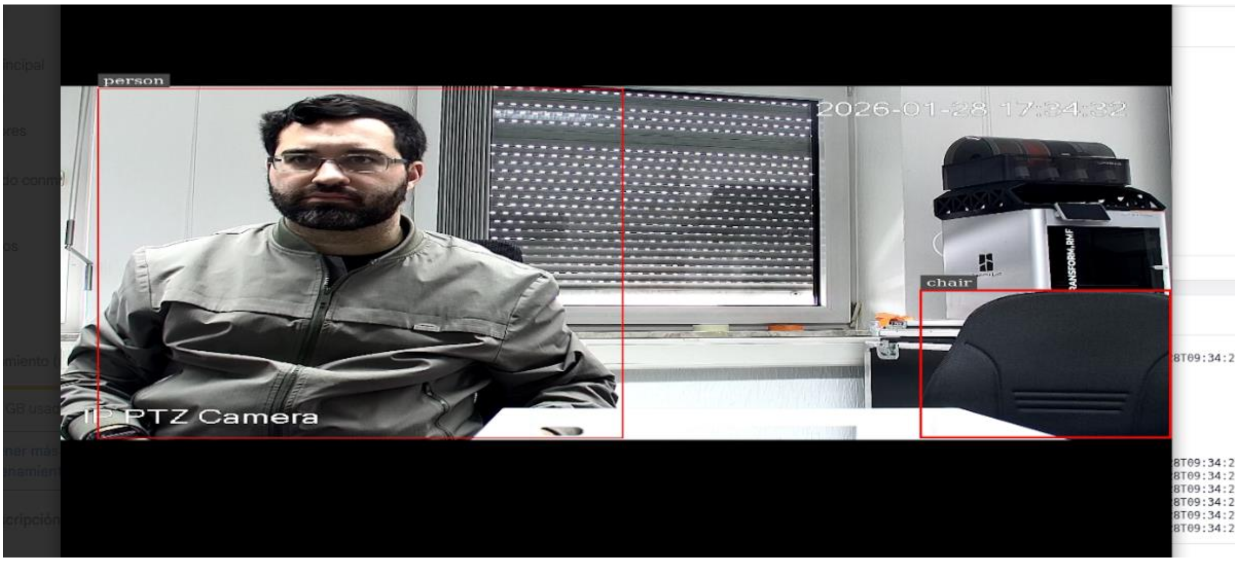


Figure 45. Detection with bounding boxes example

5.12. Use the on-screen arrow controls to move the PTZ camera.

5.13. Press the Zoom button repeatedly to the right to zoom in, and to the left to zoom out.

5.14. While moving the camera, the detections should be displayed on the monitor with bounding boxes and the object name, and the relay's red light should either flash or turn on, depending on the detections. The ultrasound should start beeping intermittently or permanently.

5.15. Press Stop to close the window. It should close after a few seconds

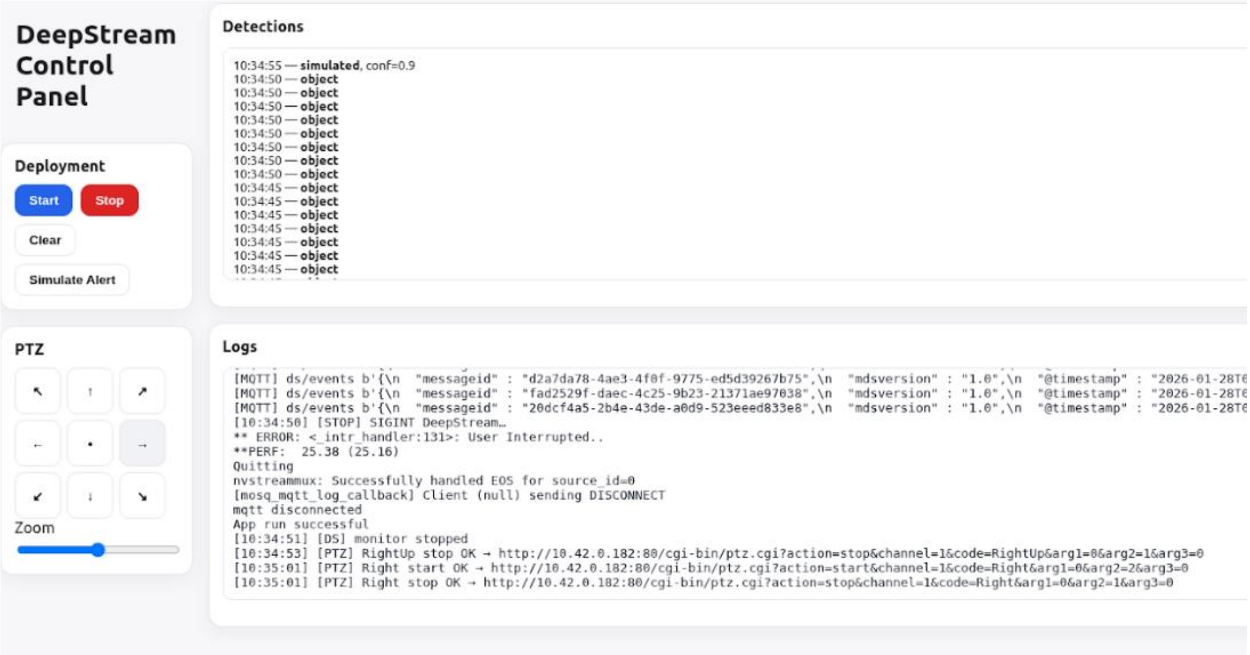


Figure 46. Panel logs

5.16. Press Clear to reset the logs
5.17. Press Simulate Alert to test the ultrasound
5.18. To completely stop the program, close the URL and then press CTRL-C on the terminal as displayed in Figure 47 or simply close the terminal.

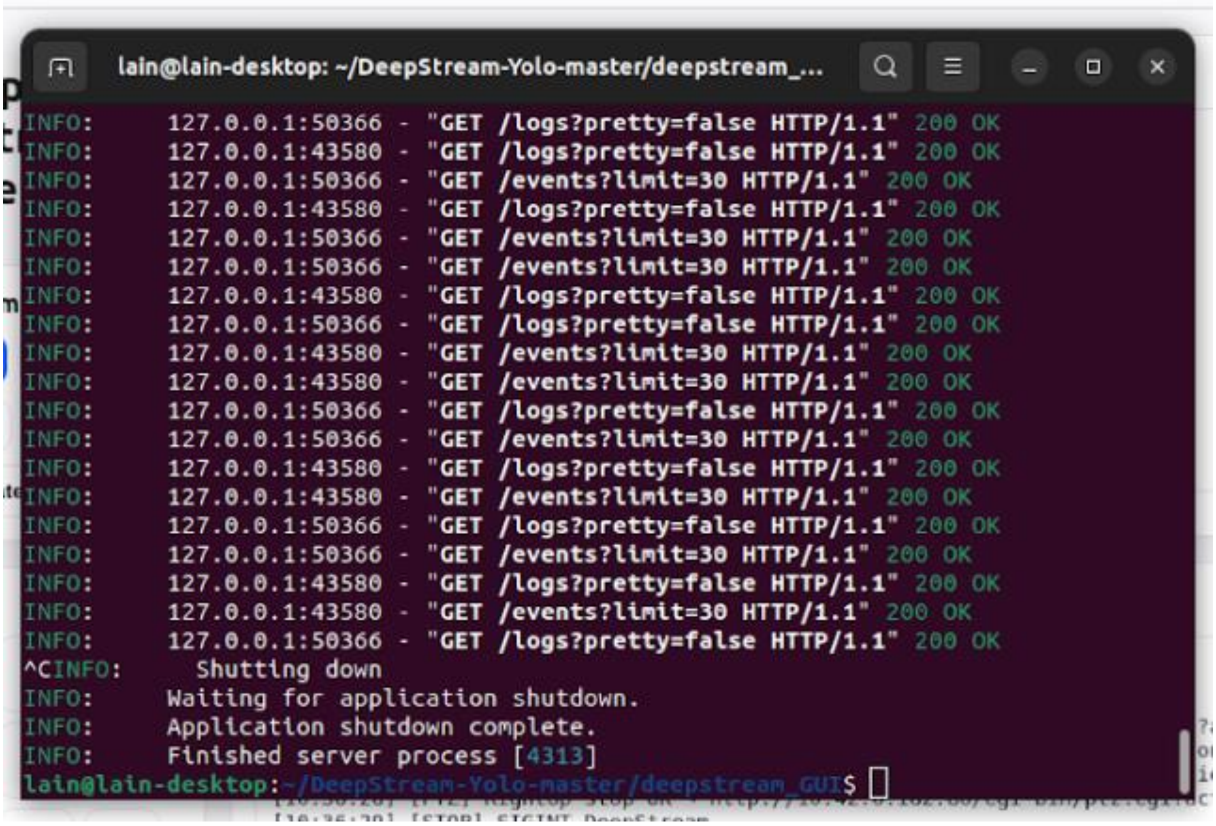


Figure 47. Process termination with the terminal

If any problems arise at this stage, a backup GUI similar to the main one, that uses a USB camera (also included), can be executed for at least testing the entire pipeline. For this method, the only requirement is internet connectivity on the Jetson, prefer via Wi-Fi. Follow these steps to use this GUI.

Back-up GUI:

1. In the folder DeepStream-Yolo-Master click on USB_GUI_Backup. Figure 48 shows this directory.

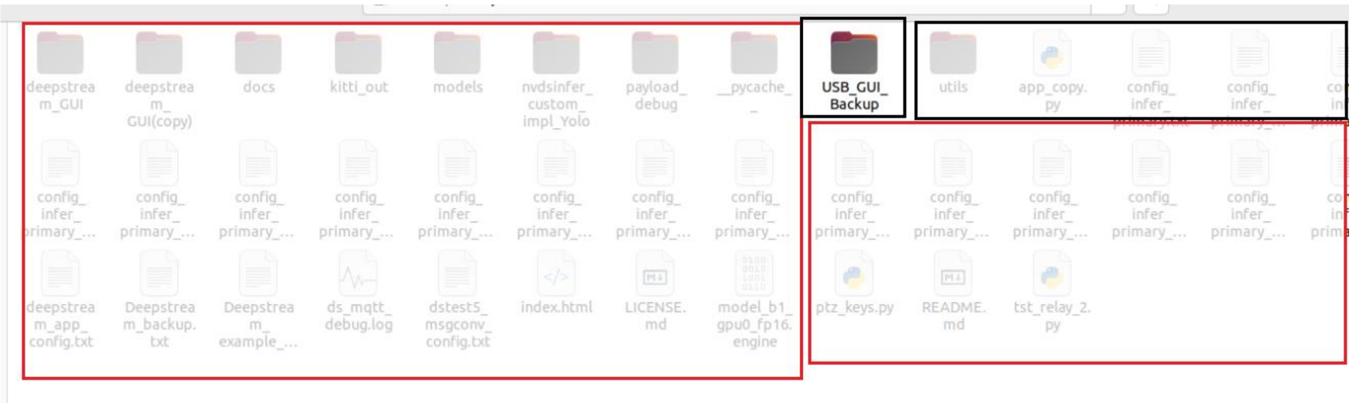


Figure 48. USB backup folder

2. Open a new terminal from this folder

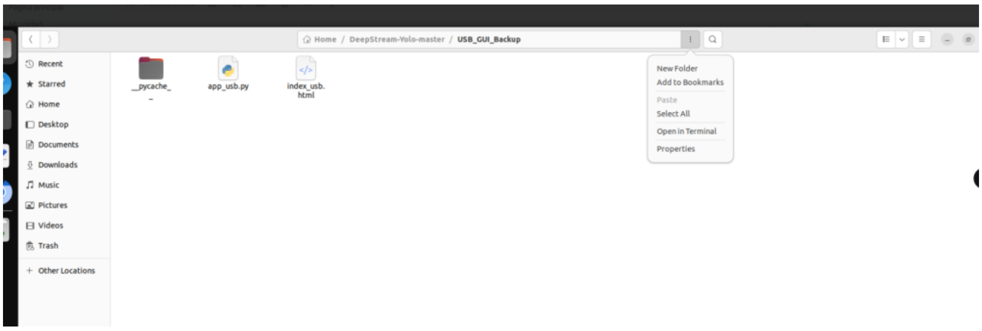


Figure 49. Backup API folder

3. Connect a USB camera to the Jetson. If all the slots are taken, try connecting the relay's USB port to a USB adapter to replace the camera's USB port.
4. Insert the following command: uvicorn app: app --host 0.0.0.0 --port 8000.

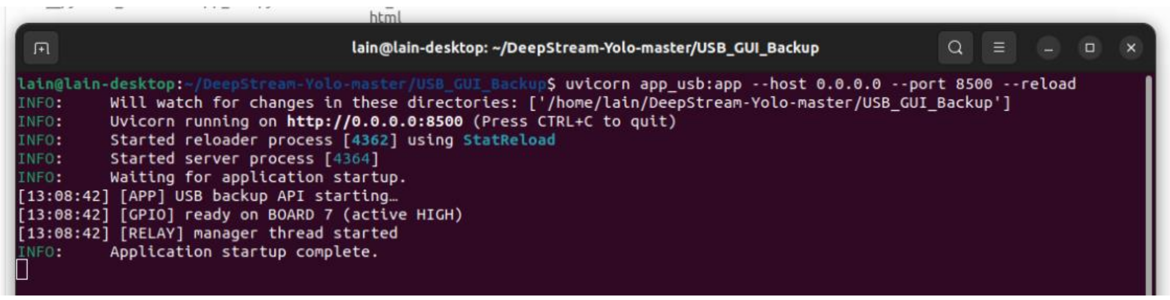


Figure 50. Uvicorn on a different port

5. Enter the URL shown in the terminal, or right-click http://0.0.0.0:8000 and open the link.
6. The GUI will appear on the screen. Press Start. A window should pop up on the screen. This configuration uses the same model as in step 5.13.

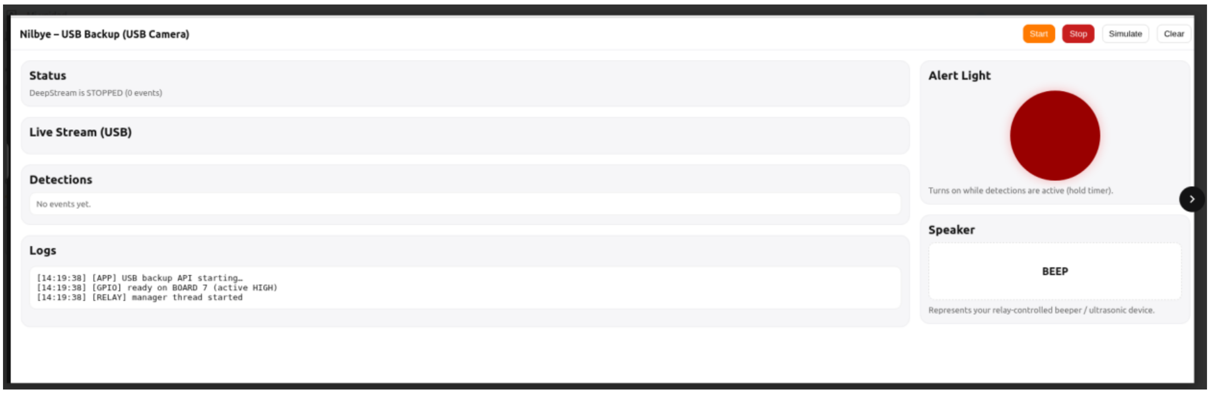


Figure 51. Backup GUI with USB

7. While moving the camera manually, the detections should be displayed on the monitor with bounding boxes and the object name, and the relay's red light should either flash or turn on, depending on the detections. The red light forms the relay should start flashing.
8. Follow the same steps from 5.17 to 5.20.

This concludes the Developer Mode. In the next phase, Headless Mode, the system would be configured for future deployment and later incorporated into the energy network.

So far, this pipeline has used a pre-YOLOv8-trained model from Ultralytics. To change to a similar model trained for Project Nilbye that has the class Nilgai go to the DeepStream-Yolo-local_test folder located also on the main folder next to Deepstresm-Yolo-master and repeat the steps from 5.7 onwards.

PHASE II: Headless Mode

In the second phase, the same GUI from the previous step would be used on a windows pc or laptop. Midway through this procedure, the NVIDIA Jetson Orin Nano would start being powered by the Litionite Venom Power Bank, as well as the relay on Level 2. That means that by the end of these configurations, the peripheral connections for monitor, mouse, and keyboard would be removed. For better control while setting Headless Mode, the PTZ Camera would still be powered by a fixed power source.

6. Install and configure Tailscale. This program would enable a headless boot, allowing control of the Jetson without a monitor and from another computer.
    6.1. On the Windows pc or laptop, download the Tailscale app from https://tailscale.com/download. As previously outlined in developer mode, this PC or laptop must be connected to the same network as the Jetson.
    6.2. Install the Tailscale app
    6.3. Open the app
    6.4. Log in with an account for Google / Microsoft / GitHub / email
    6.5. Once logged in, the PC or laptop should receive its own IP address 100. x.x.x address and should be connected to the dashboard.
    6.6. On the Jetson, Tailscale is already installed to register the device; open a terminal
    6.7. Type sudo tailscale up. Tailscale would activate
    6.8. The terminal would print a log in URL for Tailscale. Back on Windows, log in to Tailscale again, or refresh the Tailscale dashboard if it is still open. The NVIDIA would get an IP 100. x.x.x address. Figure 52 shows a typical example of connections listed on the Tailscale dashboard,

MACHINE	ADDRESSES	VERSION	LAST SEEN
lain-desktop horacioserrano1989@gmail.com SSH	100.89.109.90	1.92.5 Linux 5.15.148-tegra	Connected
laptop-26dsha4o horacioserrano1989@gmail.com	100.77.197.54	1.90.9 Windows 10 22H2	Connected

Figure 52. Tailscale Dashboard

7. Enable auto-start and full headless.
    7.1. On the Jetson, open the folder from 5.6 and click on the file Deepstream_example_usb.txt. The file is on the bottom right corner as seen in Figure 53.

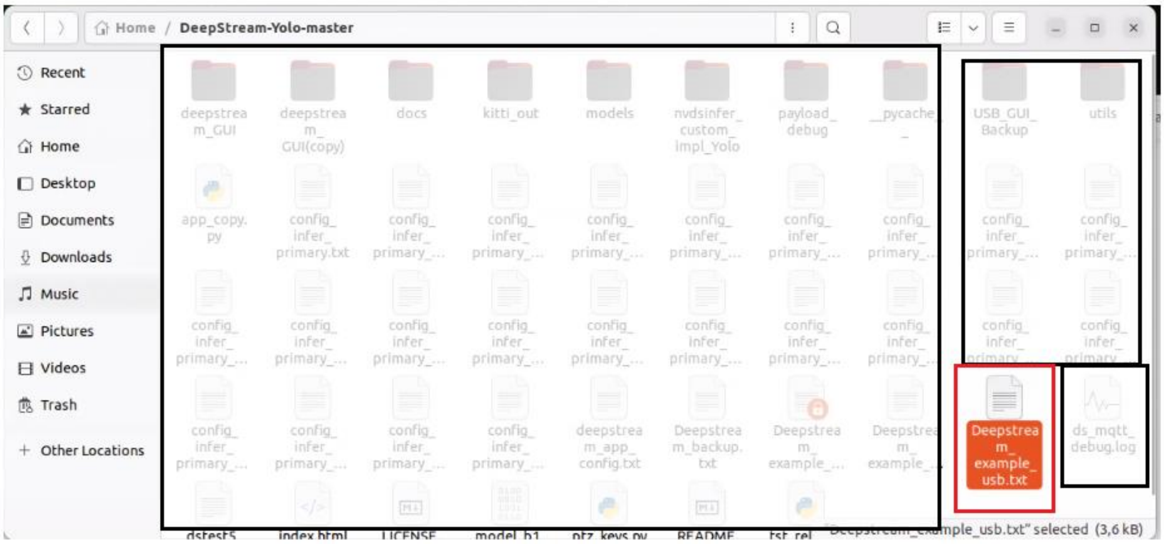


Figure 53. Deepstream_example_usb.txt file on DeepsStream directory

7.2. Disable the [sink0] segment by replacing enable=1 with enable=0. Save and close the file.

The section is between the 36 and 45 lines as seen in Figure 54. **NOTE:** Without disabling this sink, the camera won´t be able to detect objects.

```
35
36 [sink0]
37 enable=0
38 #Type - 1=FakeSink 2=EglSink/nv3dsink(Jetson only) 3=File 4=RTSPStreaming 5=nvdrmvideosink
39 type=2
40 sync=0
41 conn-id=0
42 width=0
43 height=0
44 plane-id=1
45 source-id=0
46
```

Figure 54. Toggling of sink0 on Deepstream file

7.3. Open a terminal and run the following command: sudo systemctl enable nilbye.service. This action configures the system to start the API automatically on every boot.

7.4. Verify the system configuration by typing on a terminal sudo systemctl status nilbye.service --no--pager. The service should output Active, meaning that it exists and it´s working. The active green sign should be display as in Figure 55.

```
lain@lain-desktop:~$ sudo systemctl enable nilbye.service
[sudo] password for lain:
lain@lain-desktop:~$ sudo systemctl status nilbye.service --no-pager.
systemctl: unrecognized option '--no-pager.'
lain@lain-desktop:~$ sudo systemctl status nilbye.serv
lain@lain-desktop:~$ sudo systemctl status nilbye.service --no-pager
● nilbye.service - Nilbye API (FastAPI/Uvicorn)
     Loaded: loaded (/etc/systemd/system/nilbye.service; enabled; vendor preset: enabl
     Active: active (running) since Wed 2026-02-04 11:44:36 CET; 2h 21min ago
   Main PID: 1124 (python3)
      Tasks: 3 (limit: 8809)
     Memory: 64.8M
        CPU: 12.440s
     CGroup: /system.slice/nilbye.service
             └─1124 /usr/bin/python3 -m uvicorn app:app --host 0.0.0.0 --port 8…

Feb 04 11:44:36 lain-desktop systemd[1]: Starting Nilbye API (FastAPI/Uvico…)...
Feb 04 11:44:36 lain-desktop systemd[1]: Started Nilbye API (FastAPI/Uvicorn).
Feb 04 11:44:43 lain-desktop python3[1124]: [GPIO] ready on BOARD 7 (active …GH)
Feb 04 11:44:43 lain-desktop python3[1124]: [11:44:43] [MQTT] Connecting to…883…
Feb 04 11:44:43 lain-desktop python3[1124]: [11:44:43] [MQTT] Connected 127.…883
Feb 04 11:44:43 lain-desktop python3[1124]: [11:44:43] [MQTT] Subscribed 'ds…ts'
Feb 04 11:44:43 lain-desktop python3[1124]: INFO:     Started server process…24]
Feb 04 11:44:43 lain-desktop python3[1124]: INFO:     Waiting for applicatio…up.
Feb 04 11:44:43 lain-desktop python3[1124]: INFO:     Application startup co…te.
Feb 04 11:44:43 lain-desktop python3[1124]: INFO:     Uvicorn running on htt…it)
Hint: Some lines were ellipsized, use -l to show in full.
```

Figure 55. A similar window is expected when enabling the service

7.5. In case the service didn´t start, it can be run manually with sudo systemctl start nilbye.service. Repeat step 7.2 to verify.
7.6. Restart the Jetson
7.7. Unplug the HDMI cable, mouse, and keyboard
7.8. Wait between 60 and 90 seconds for the Jetson to boot again.
7.9. On a PC or laptop running Windows, open a browser and go to http://<JETSON_TAILSCALE_IP>:8000 or http://<JETSON_LOCAL_IP>:8000. You will see the same GUI as in developer mode and can control the system as in steps 5.13 to 5.19.
7.10. To shut down the system properly, go to PowerShell
7.11. Type ssh -v root@<JETSON_TAILSCALE_IP>. The NVIDIA can be controlled via ssh this way. PowerShell should show the user as root.
7.12. Still on PowerShell, type: sudo shutdown now

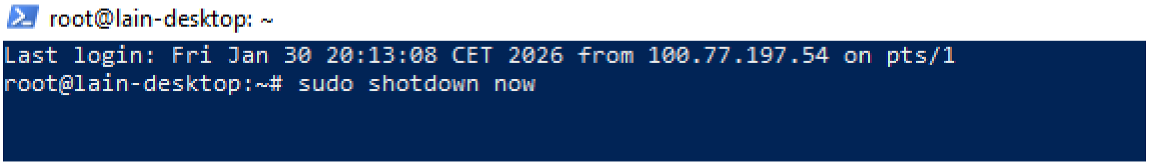


Figure 56. Remote use with PowerShell

7.13. The system is ready to start headless once booted, and the Jetson can be controlled via SSH without a monitor. To stop the service and return to Developer Mode, either via SSH or with a monitor, open a terminal and type sudo systemctl stop nilbye. service. Afterwards, go back to the file from 7.1 and enable the [sink0]. NOTE: Without enabling this sink and not stopping the service, the camera won´t be able to detect objects in Developer Mode again.

PHASE III: Integration with Energy Network

In the last phase, the existing set-up from the previous steps would be integrated into a solar-powered energy network. Meaning, that the Litionite power bank itself powers the other components in the case, and the Dahua PTZ camera would be powered now independently from any external fixed power source.

8.1 Place the Enjoy Solar panel between 30 and 80 cm above ground on a 45 ° angle facing the sun and avoiding shadow.

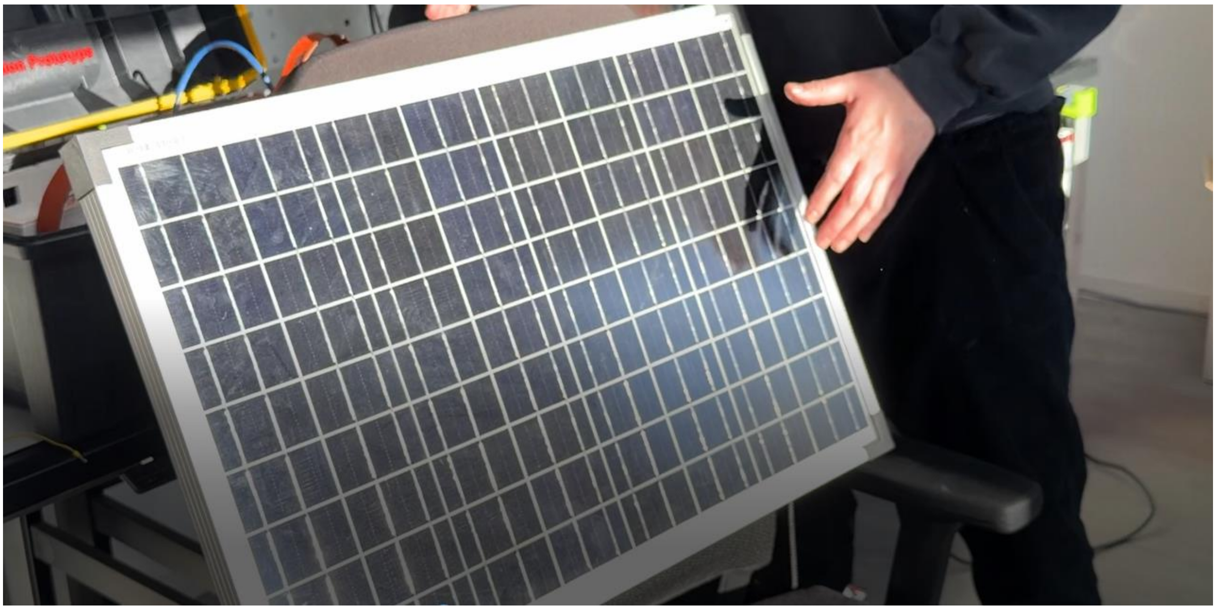


Figure 57. Example placing of Enjoy Solar Panel

NOTE: The current item list supports the aforementioned panel setup. However, if a higher placement height is desired, please consider procuring longer cables than those provided to ensure proper system operation.

8.2 Open the toolbox, with the leather belts logo facing front and the powerbank PB label on the right position as shown on Figure 58. This box contains all the necessary connections to power the energy grid. **WARNING:** Connect only the mentioned wires and do not touch the ones that are not mentioned on the following instructions.

NOTE: If any problems or irregularities with the energy network connections are found, the entire array can be retrieved from the toolbox, given the initial state found on Figure 58.

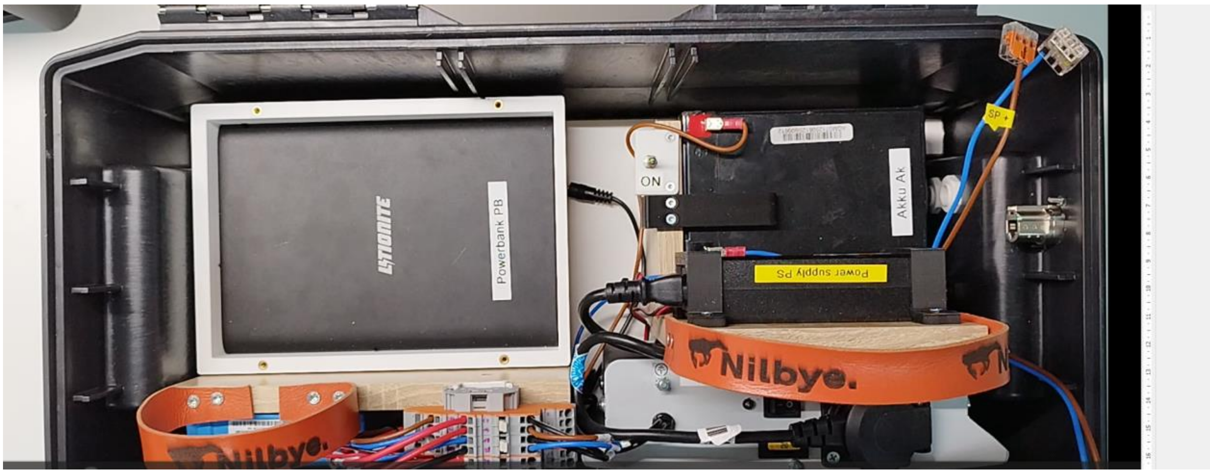


Figure 58. Open toolbox

8.3 Grab the black and red MC4 solar photovoltaic connectors; They are labeled as PV cables. Insert the black connector on the lower DC socket next to the Greencell 12V battery labeled as Akku Ak on the right side and then the red connector on the upper DC socket

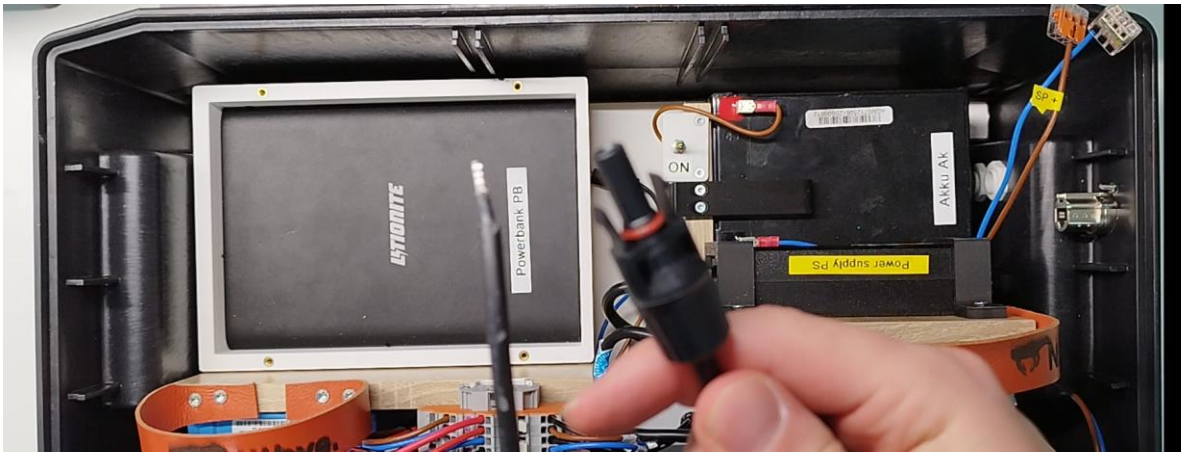


Figure 59. Black MCA solar photovoltaic connector over the toolbox

8.4 Tighten both DC sockets screw outside of the toolbox

8.5 Grab the Cam + and Cam- cables that correspond to the PTZ camera and connect them to the DIN rail terminal block as seen on Figure 60 and 61.

Figure 60. Cam + and Cam - cables

Figure 61. Both cables connect on the DIN terminal

8.6 Connect the two PV cables to the DIN rail´s positive and negative terminal as shown on Figure 62.

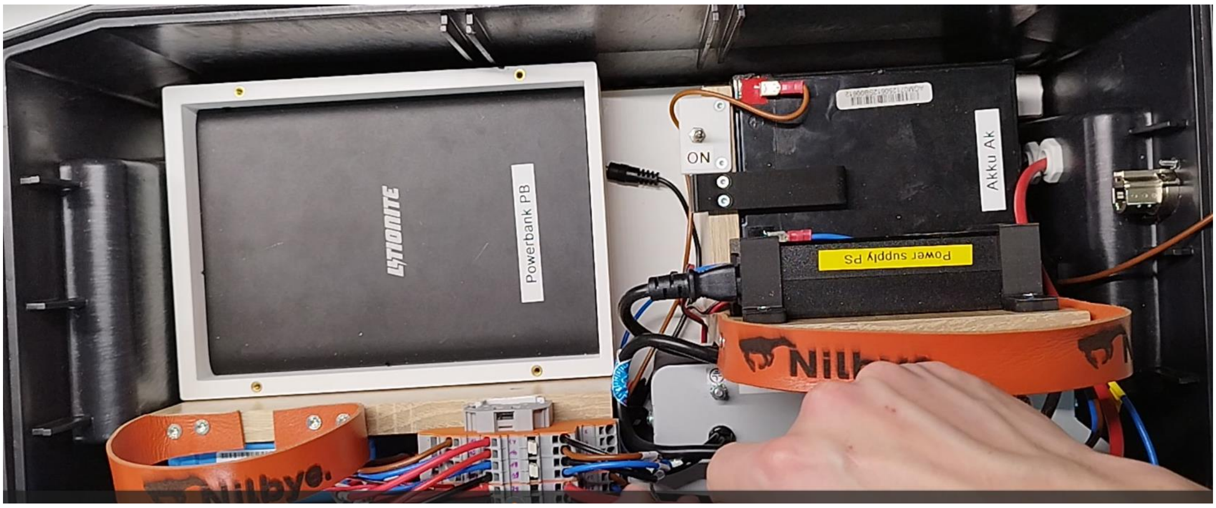


Figure 62. Jack cable disconnected

8.7 Connect the small jack cable on the toolbox floor to the powerbank on the IN port. The connection should be exactly as shown on image.

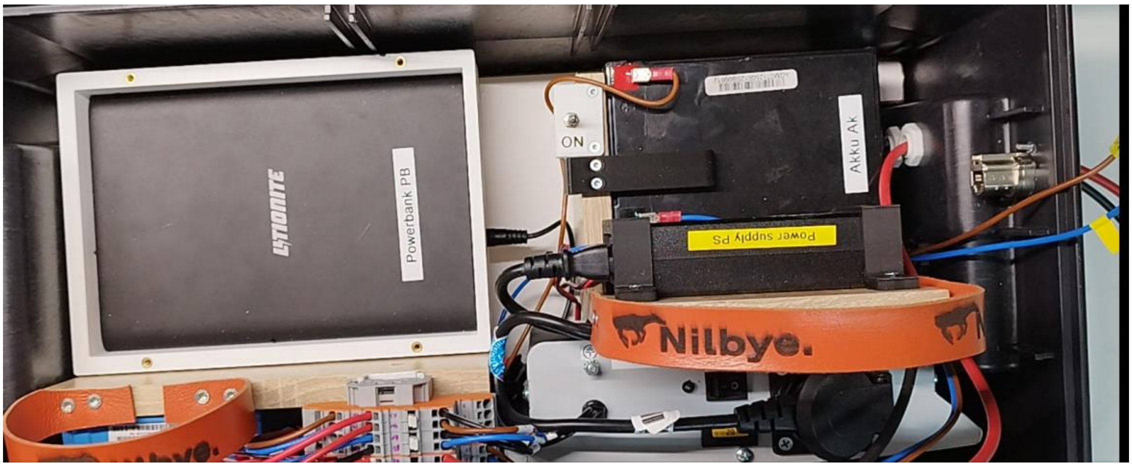


Figure 63. Cable connected

WARNING: Do not use the DC-OUT 16/20/24 V or 12V ports for this step, because those are the power bank outlets. For this step use only the IN for receiving power.

Figure 64. IN port next to powerbank display

8.8 Grab Level 1 and place it over Level 0. Then connect the ethernet cable to the proper socket locate on the right side above the DC sockets.

Figure 65. Level 1 integrated to the energy grid

8.9 Bolt the two planes together

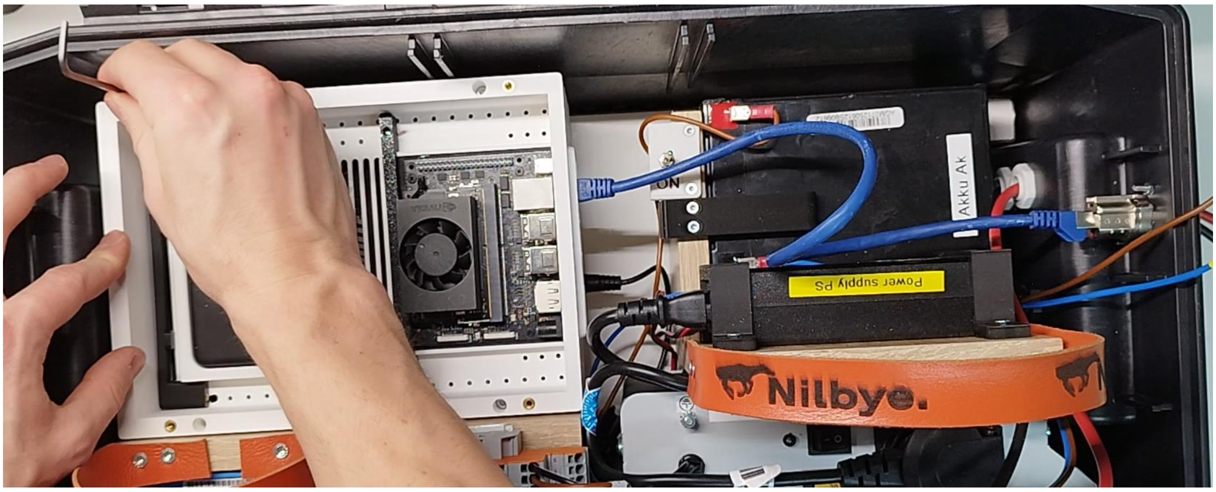


Figure 66. Mounting of levels

8.10 Repeat the steps 3.1 to 3.3 from the Developer Mode to mount all the levels together.
8.11 From Level 2 grab the cable with the negative (black) and positive (red) ends and inserted them on the Wago connectors correspondent to SP- and SP+ cables.

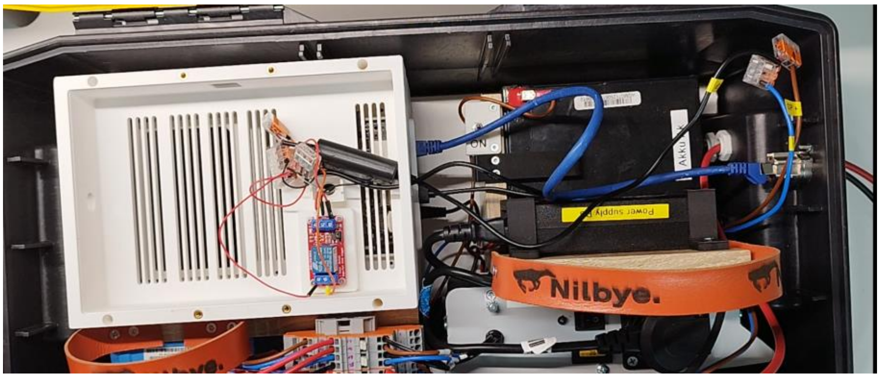


Figure 67. Integration of Level 2 to the energy grid

8.12 Place the lid over Level 2 and bolted it to enclose the main case

Figure 68. Enclosing of the main case on the toolbox

8.13 Grab the jack cable shown on Figure 69 and insert it on the 12V DC-OUT port on the powerbank, to power the Jetson. The image shows an example of how this connection must take place. The cable has an attached adapter to one end, for the IN port but this end can also be used for the Jetson input. **WARNING:** Do not use the DC-OUT 16/20/24 V or IN ports for this step.

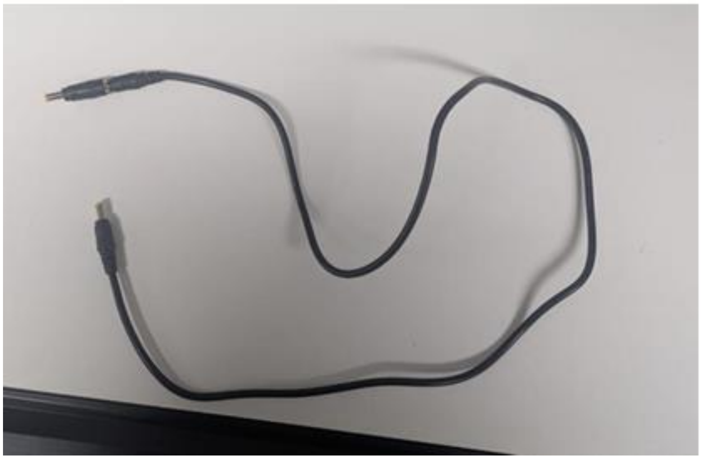


Figure 69. Powering cable for the Jetson

Figure 70. Example of powering on the main case using the Litionite Venom

8.14 On the exterior right side of the toolbox, open the upper IEC appliance coupler with two inputs and connect the cable wired to the ultrasound.

Figure 71. Ultrasound with respective cable

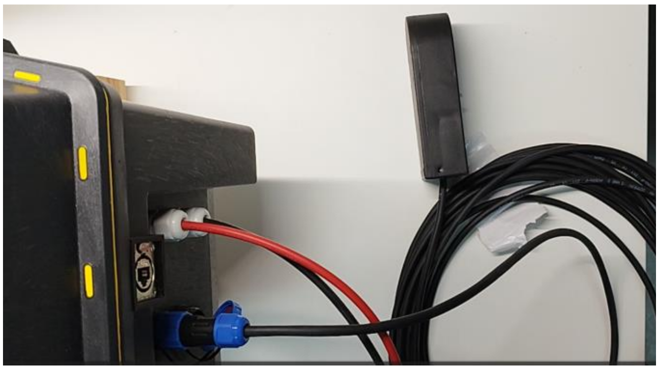


Figure 72. Connecting the ultrasound to the toolbox

8.15 Repeat the same process for the 3-pin cable that powers the PTZ camera.

Figure 73. Camera cable connected to the toolbox

8.16 Grab the ethernet cable and plug it in into the respective socket. Then connects the cable to the camera´s ethernet port. Figure 75 shows an example of this setting but usually the camera should be mounted.

Figure 74. Ethernet cable connected to camera

Figure 75. Example of typical camera mounting

8.17 Plug in the MC4 connectors with the respective solar panel cables placed on the back. Each one corresponds to the same color black for negative and red for positive.

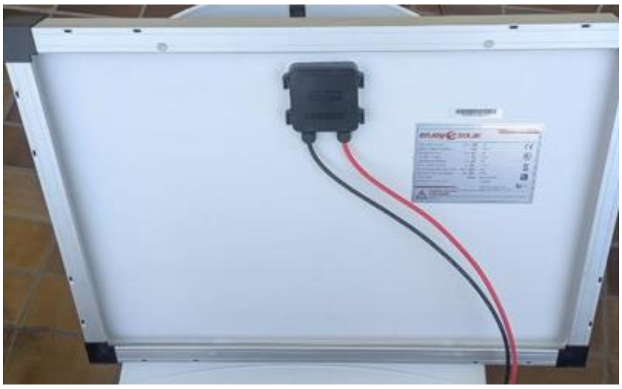


Figure 76. MC4 cables on the back of the Enjoy Solar Panel

Toggle the ON switch to power the entire energy network and turn on the Voltcraft SW-150 Inverter.

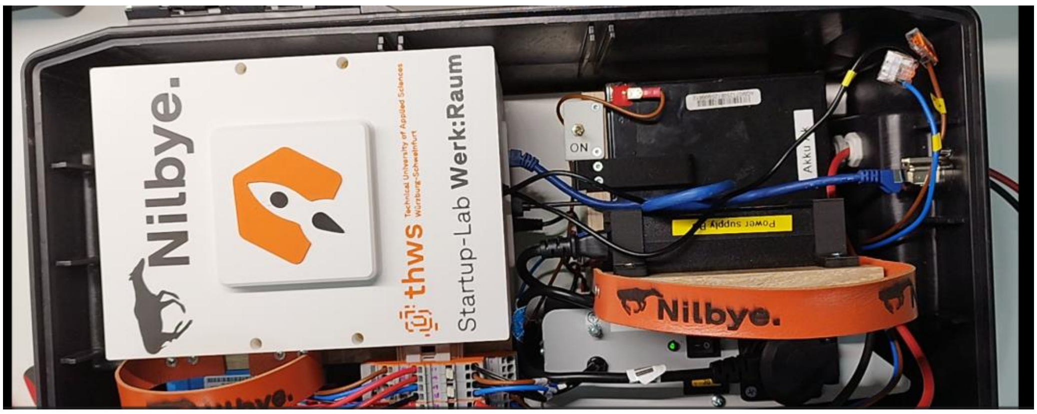


Figure 77. Energy network fully operational

8.18 At this point, the blue light on the Victron next to Bulk should be flashing on the Victron SmartSolar MPPT controller. If there is an ideal amount of sunlight and the solar panel is not shaded, the Victron should start displaying lights next to Absorption, then Float. Ideally, wait for the network to reach this last status before turning on the powerbank. All the necessary connections are detailed on page 29.

NOTE: Load outputs are reserved for low-consumption devices; they are not intended for charging prototype devices.

8.19 Before continuing, download the Victron Connect app for Android or iOS on your mobile phone. Enable Bluetooth to connect to the Victron. This app will display panel and battery metrics, with the most important being the number of Watts produced as shown on Figure 79 . With it is also possible to monitor the current state of the controller. The PIN is 449543 and is located on the back of the device.

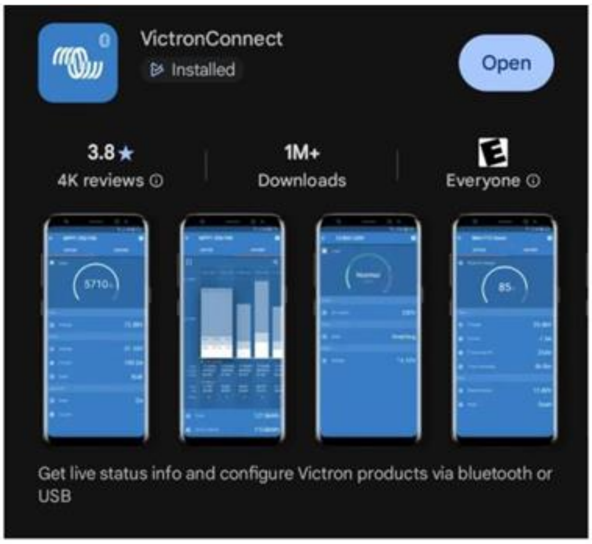


Figure 78. VictronConnect app

Figure 79. Example of app metrics

Nr	Name		Name	Nr
1	MPPT Load +			2
3	MPPT Load -			4
5				6
7				8
9	Power Supply for Camera + (PS)		Plug Camera +	10
11	Power Supply for Camera – (PS)		Plug Camera -	12
13	MPTT Batt +		Voltcraft Inverter + (Inv)	14
15	Powerbank + (PB)		Battery + (Ak)	16
17	MPTT Batt -		Voltcraft Inverter - (Inv)	18
19	Powerbank - (PB)		Battery – (Ak)	20
21	MPPT PV +		Cable PV + (red)	22
23	MPPT PV -		Cable PV – (black)	24

Figure 80. Close up of terminal connections

8.20 Once float is reached, turn on the powerbank by pressing the button on the left side.

Figure 81. Power button on left side of the powerbank

Proposed Future Streamlining

With the completion of these phases, the system should be operational and ready for deployment. The system can be arranged in different configurations depending on the area, and it can also be modified to suit the desired application and outcome.

The aforementioned steps are intended for a first configuration-type scenario. Once successfully executed, future deployments would only need the toolbox and main case, which contains all the backbone and peripheral components, to be connected to the camera and the solar energy panel. To this end, network connectivity would be configured in advance, allowing SSH access via an API and API control all in Headless Mode with auto-start enabled.

This streamlining would eliminate the internet and camera configurations from Phase I, along with the arming of the case, and would integrate the remaining steps with Phase II. Given that Phase III is complete and the camera is connected to the energy network before deployment, future users will only need internet credentials to access the API.

The projected scheme for future deployments would encompass a structure in which the camera is mounted. Next to it, the solar panel is placed on a high point over the surface. These two elements are then connected to an enclose energy network that contains the main case. This arrangement significantly simplifies the connections, allowing a more efficient deployment.

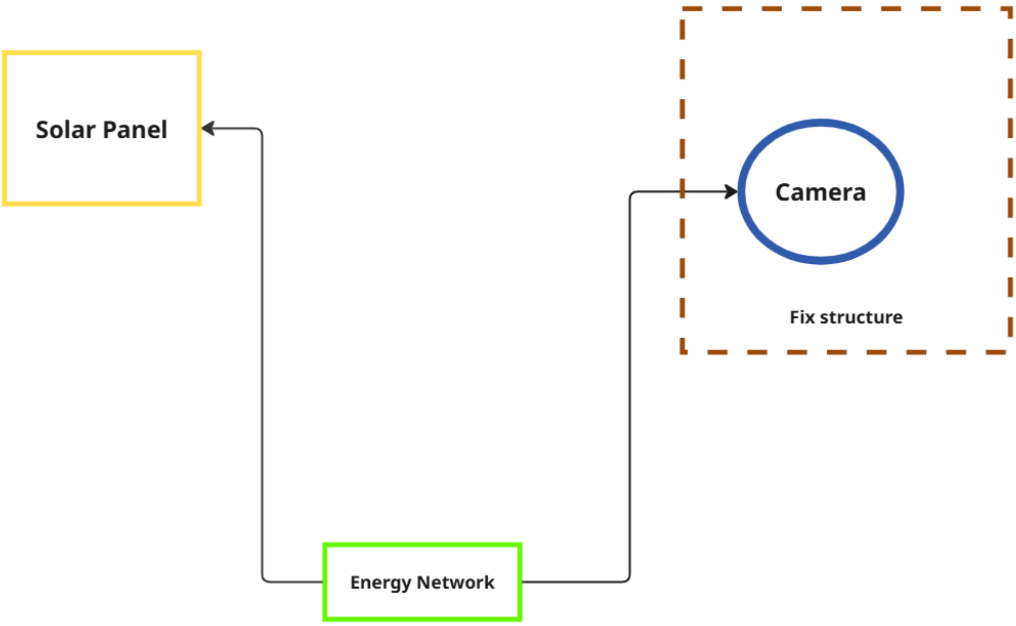


Figure 82. Propose future array

Further documentation and updates from Project Nilbye can be found on the GitHub repository *https://github.com/hbo-bomb/Project-Nilbye.*

For any technical support or questions regarding the Prototype or Project Nilbye itself and access to the project repository, please don´t hesitate to contact *horacioserrano1989@gmail.com*. For the access attached your GitHub username or email to get an invite.